

Tide retrospective: turnkey identifiability

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

2026-08-09

1 Setting

Eighth tide of the germbij arc, removing its last non-mathematical hypotheses. The analytic corollary of the previous tide¹ still carried three per- t integrability premises. For continuous potentials and a continuous compactly supported bump these are automatic, and this tide makes them so, yielding the arc’s headline statement in its final form.

2 The thing formalised

Four declarations in `Laplace/Turnkey.lean`, all sorry-free with zero warnings. Three integrability lemmas: the minorant and each pencil slice are continuous with support inside $\text{tsupp } \psi$, hence integrable; and the uncurried pencil integrand is integrable on $[0, 1] \times \mathbb{R}$ with the restricted product measure, by a cylinder-indicator majorant. Then the turnkey theorem²: for $L_1, L_2 \geq 0$ analytic at 0 with differing germs, globally continuous, with $L_1 + L_2 \leq C_0 w^2$ on $[0, R]$, and ψ continuous, compactly supported, nonnegative, $\equiv 1$ on $[0, R]$: there exist m (the vanishing order of the difference), $c > 0$, $r_0 \in (0, R]$ with

$$c \cdot t \cdot t^{-m-1/2} \leq \int (L_2 - L_1) \psi (e^{-tL_1} - e^{-tL_2}) dw \quad \text{for all } t \geq 4/r_0^2.$$

Every hypothesis is now a mathematical property of the potentials and the bump; nothing about integrals survives in the statement. This is the 1D germbij Theorem 7.3 chain in the cleanest form the note could cite.

3 Proof strategy

The two one-dimensional lemmas are `Continuous.integrable_of_hasCompactSupport` with the support transported from ψ (`HasCompactSupport.intro` on $\text{tsupp } \psi$; off the support the integrand vanishes because ψ does). The product lemma follows the consult’s blueprint exactly: joint continuity by `fun_prop`; a uniform bound on the compact cylinder $[0, 1] \times \text{tsupp } \psi$; the indicator majorant on $\text{univ} \times \text{tsupp } \psi$, integrable because the restricted product measure of the cylinder is finite; the restricted product rewritten as a restriction of the full product to extract almost sure membership of the s -coordinate in $(0, 1]$; and domination. The turnkey theorem is then the analytic corollary fed by the three lemmas.

¹2026-08-09-03-25-tide-germbij-analytic-corollary.

²analytic.pencil.difference.lower_bound’.

4 Roadblocks and resolutions

Two elaboration-level snags. `isCompact_Icc`'s implicit endpoints unified with an ambient variable named t rather than $[0, 1]$, producing a bound over the degenerate interval $[t, t]$; explicit `(a := 0) (b := 1)` fixed it. And an anonymous-constructor membership proof inside a `rw` argument could not determine its type; naming it in a `have` first resolved it. The consult was used tactically this time (idiom-level, before writing the product lemma) and its recipe compiled with only those two repairs.

5 What was Mathlib, what was new

All Mathlib: the compact-support integrability lemma, the compact-bound lemma, the product-restriction identity, indicator integrability, and domination. New: only the packaging. The arc's quantitative content was finished three tides ago; these last tides have been about making the Lean statement coincide with the note's.

6 Lessons

Beware implicit interval endpoints when an ambient variable could unify with them; instantiate `isCompact_Icc` explicitly in any context with free real variables. Consults are cheapest when asked for idioms before writing, not diagnoses after.

7 Follow-ups

- The multivariate leading-part instantiation (produce the scaled set S from a nonzero homogeneous leading part), the one remaining ingredient for a turnkey multivariate statement.
- Note annotation: tag Theorem 7.3 with the turnkey theorem once merged; it certifies the note's hypothesis set most faithfully.
- Consider upstreaming `integrable_pencil_product`'s pattern (continuous kernel times compactly supported factor on a restricted product) to lean-common on second use.